Course Description

Department of Computer Science and Engineering
Shanghai Jiao Tong University

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1 Engineering

EI901: Science and Technology Innovation (Part 1)

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
2.0	2.4	16 weeks	lab: 90

Science and Technology Innovation (Part 1) is a general practice course for all undergraduates in the university. The course guides students to produce a microcontroller-based electric car in a team. The car should be able to follow a predetermined curve with several sharp turns and reversals. Students will learn fundamentals of the C programming language and apply it to their projects. They will also learn the high-level architecture of embedded systems so that they could tailor their codes to limited memory and real-life complications. Through this course, students will acquire necessary programming skills in C, especially for controlling hardware motions.

EI228: Science and Technology Innovation (Part 2B)

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
2.0	2.4	16 weeks	lab: 90

Science and Technology Innovation (Part 2B) is one of the second stage course modules. The students are required to design a car model that traces the line on a map. The cars should use bluetooth module to communicate data with host computers. The project is a practical engineering subject with low complexity and fit for beginners. Students should carefully keep trade-offs between the hardware and the software complexity, the whole and the partial performance.

The students will implement their projects in a group of three members. Grading for engineering prototypes makes up approximately 65% of final grade for each group, while those for participation in ordinary activities and design reports make up 10% and 25% respectively.

EI312: Science and Technology Innovation (Part 3C)

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
2.0	2.4	16 weeks	lab: 90

Science and Technology Innovation (Part 3C) is in the intermediate stage of the course series, mainly for the senior students. The course asks students to design an electronic car system controlled by Android smartphones. In particular, students will write Android applications in Java that controls the car remotely and receives motion pictures from the car. Students should tackle technical problems and choose the correct protocols for communication. The participants are organized by groups to finish the project. Groups are graded according to completeness and innovation of their work.

EI401: Science and Technology Innovation (Part 4C)

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
2.0	2.4	16 weeks	lab: 90

Science and Technology Innovation (Part 4C) is in the ultimate stage of the course series. This year's course asks students to implement an online circuit experiment portal. The portal should integrate an account page, a circuit simulator with specific experiment tasks, and an autograder. Users can log in the portal through university's central account system and perform online experiments.

The class is divided into three cooperating teams, each in charge of a component of the system. Members are graded according to the team's accomplishment and their contribution to the team.

ME116: Introduction to Engineering

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lab: 135

This course introduces elements of engineering to beginners. For the first half of the course, students are taught principles in programming, engineering and manufacturing process. In particular, students will learn the interactions between program and hardware using the Arduino toolkit. They will build their own circuits and experiment with codes. The teaching is mostly hands-on experience.

For the second half of the course, students form groups of 3-4 members and carry out an engineering project, using the taught knowledge as guidance. The course does not impose limitations on possible topics, so students can propose any doable projects according to their interests. From design to implementation, the groups should finish and improve their proposed project by the end of the semester and give presentations to peers and instructor.

ME123: Manufacturing Practice

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
4.0	4.8	16 weeks	practice: 360

Engineering practice is a practical basic course about mechanical manufacturing technology, which is an important link to cultivate practical ability and to combine theory with practice. Through the practical teaching, the students can contact the mechanical manufacturing production technology, and have certain engineering consciousness and manufacturing practical ability. In the process of teaching, we emphasize hands-on learning via combining theory teaching and demonstration and explanation. Students are required to operate machines and tools themselves to manufacture products.

2 Mathematics

MA077: Linear Algebra

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

This is an introductory course to linear algebra. It provides students with basic concepts, principle and operating methods through in-class teaching, extracurricular teach-

ing, exercises and exams. It improves students' ability to analyze and solve problems. The outline of this course includes the following: determinants, matrices, vector and equations, linear space, eigenvalues and eigenvectors, similar diagonal matrices, real quadratic form, and linear transformation.

MA080: Calculus I

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
6.0	7.2	16 weeks	lecture: 270

The calculus course is designed to teach in two semesters. The first half of this course covers the fundamental concepts, theorems and tools of single-variable calculus. It provides not only necessary mathematical knowledge to follow-up science and engineering courses, but also proper training in mathematical abstractness, logic and tightness.

Contents include limits, continuity, derivative and differentiation, integration, and differential equations.

MA081: Calculus II

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
4.0	4.8	16 weeks	lecture: 180

The second half of the course naturally generalises the tools in single-variable calculus to multi-variable case, and builds up the connection with analytic geometry. Contents include partial derivatives, multiple integrals, line integrals and surface integrals. It also contains a chapter on series.

MA119: Probability and Statistics

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

The course is a foundation subject for undergraduates. The teaching activities included lectures, readings, assignments, homework and exams. The course includes introduction to the theory of probability and statistical analysis, general probability and conditional probability, continuous and discrete random variables and their probability distributions, expectation and variance of random variables, distribution and correlation of multiple random vectors, law of large numbers and central limit theorem, statistical estimation, regression analysis and hypothesis test.

The course aims at providing students with the basic concepts, principles of probability and statistics, as well as developing the students' ability to analyses and to solve stochastic problems.

MA238: Discrete Mathematics

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

The aim of this course is to illustrate the basic concepts, theory and methods in various branches of discrete mathematics. It also sets up the ability of abstraction, logical reasoning and conclusion construction for students.

The content covers basics of mathematical logic, set theory and graph theory. The mathematical logic part requires the students to grasp propositional and first-order logic, and to do reasoning by these tools. The set theory part covers the standard notations for a set, and builds relations and functions based on it. Finally, the graph theory part introduces various kinds of graphs, the notion of connectivity, Eulerian circuits and Hamiltonian cycles, and trees.

3 Physics

PH001: Physics I

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
4.0	4.8	16 weeks	lecture: 120

Physics is one of the basic courses for students who will be scientists and engineers in future. The goal of this course is to help students understand the fundamental physical laws and know how these laws can be applied to solve many problems and how physics is relevant to modern applications in the world around them.

The course is taught in two semesters (i.e. one year). Students will learn classical mechanics, electromagnetism, thermodynamics, physical optics and modern physics systemically in the year. The first half of the course introduces motion, Newton's mechanics, energy and momentum, rigid body and rotation, vibration and waves, and thermodynamics.

PH002: Physics II

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
4.0	4.8	16 weeks	lecture: 120

The second half of the physics course introduces electric field, magnetic field, Lorentz's force, Faraday's law and Ampere's law, Maxwell's equations and electromagnetic wave, light and its wave-particle duality, reflection and refraction in electromagnetic framework, and finally, quantum mechanics basics.

PH028: Physics Lab. I

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
1.0	1.2	8 weeks	lab: 120

The course is set up to make students hold the ideas, the fundamental principles and the basic methods of the physical experiments. It teaches students the way to use basic experimental instruments and apparatus, and the way to deal with data. The course is set up also to promote students' ability to research.

PH029: Physics Lab. II

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
1.0	1.2	8 weeks	lab: 120

Through the learning of this course, students should review and reinforce the understandings about operations of instrument, realization of experiments and analysis of data. They should know about the underlying motivation of designing physical experiments, and be able to design reasonable schemes based on the goal of experiment as well as to determine the parameters. They should acquire fundamental methodologies to analyze errors and evaluate results.

EI111: Circuit Theory

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

The course introduces fundamentals of circuit analysis. Through this course, students learn the abstraction of electronic devices and the key methods to analyze them. Course contents include: analysis of static linear circuits, operational amplifiers, dynamic RLC circuits, and sinusoidal steady states.

EM210: Theoretical Mechanics

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
4.0	4.8	16 weeks	lecture: 180

The course introduces theory of analyzing complex mechanical system. It is a step further from the physics course by stressing specialized tools in classical mechanics. The course will handle both static and dynamic cases, and allow students to apply these techniques to a variety of practical models.

4 Computer Science

CS902: Thinking and Approach of Programming

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135 lab: 120

This course is for first-year students with little or no programming experience. It helps students understand the role that computation plays in problem-solving. It also aims at establishing their ability to write small programs to solve practical problems. The class will teach the C++ programming language, from basic syntax to function encapsulation, and finally to object oriented programming. It also touches some of the simplest algorithms such as sorting and binary search.

CS214: Algorithm and Complexity

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

The course provides students a broad view of classical algorithms and analysis techniques. It also touches some cutting-edge algorithms that prepares students for future study.

Topics covered include: amortized analysis, sorting algorithms, divide-and-conquer, greedy strategy and matroids, dynamic programming, linear programming, MST algorithms, shortest path algorithms, network flow, Turing machines, reduction and NP completeness, approximation and randomised algorithms.

Textbook: *Introduction to Algorithms*, by Thomas H.Cormen, Charles E.Leiserson, Ronald L.Rivest and Clifford Stein.

CS149: Data Structure

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

This course is one of the core modules in the department of computer science. This course will introduce basic knowledge and the most important methods used in the design and analysis of computer algorithms. By finishing the course, students will: 1) know how to implement basic data structures; 2) know the relationship between data structures and algorithms; 3) be able to choose appropriate data structures and algorithms to solve a complex problem; 4) analyze simple algorithms and discuss tradeoffs among data structures.

Laboratory work is essential in this course. It stresses the development of problem solving skills through a number of programming assignments.

Contents include: linked lists, queues, stacks, binary heaps, binary search trees, balanced binary trees (AVL tree, red-black tree, AA tree and splay tree), B-trees and external sorting, hash table, union-find structure, basic graph algorithms.

EE212: Introduction to Electronics

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

This is a core course for students major in electrical engineering and computer science. It provides solid foundation for the design of digital circuits and systems. The course begins with boolean logic and Karnaugh map, and goes on to build arithmetic units, multiplexers, encoders and other basic electronic components. It then introduces modal components including latches, flip-flops, shift registers, and counters. The treatment is rigorously based on boolean logic. Through the study of this course, students are able to design simple electronic systems on their own.

CS499: Mathmatical Foundations of Computer Science

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

This course is designed for second-year undergraduates in computer science. It gathers the essential mathematical foundations for computer science study. Most of the course

will concentrate on discrete structures, but will sometimes borrow analysis techniques. It also delivers some of the highlights in combinatorics.

Course contents: combinatoric counting, famous number sequences, generating functions, probablistic methods, partial order, Mirsky's Theorem and Dilworth's Theorem, infinite sets and diagonalization, Prüffer code and Kirchhoff's matrix tree theorem, graph isomorphism and automorphism, network flow and bipartite matching, Kőnig's theorem, Hall's theorem and their application.

Textbook: *An Invitation to Discrete Mathematics*, by Jiří Matoušek and Jaroslav Nesetril.

Material: Video lectures and corresponding slides can be found at Coursera.

Grading: 50% final exam, 50% homework. The students will form groups of size 4-5. Homework will be handed in by every group.

CS467: Computing Theory

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

The topics covered in this course include: the theory of automata and formal language, computability by Turing machines and recursive functions, uncomputability, computational complexity, etc. The treatment is mathematical but the viewpoint is that of computer science. This course addresses the key problems in theoretical computer science. We intend to trigger the students' interests in theoretical computer science during the process of delivering knowledge.

Textbook: *Introduction to the Theory of Computation*, by Michael Sipser.

CS359: Computer Architecture

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

This course is an important compulsory subject of computer major. The aim is to improve the abilities of research on computer systems from levels of architecture, system analysis and performance evaluation. Through the course students can understand the basic concepts, principles, design and analysis methods of computer architecture, and know the history and development of computer architecture.

The main contents include: performance evaluation metrics, the hierarchy of memory, instruction level parallism, thread level parallism, superscalar. New technology trends, including multicore processor, GPU, and data center will be discussed in the course.

Textbook: *Computer Systems – A Programmer's Perspective*, by Randal E. Bryant and David R. O'Hallaron.

CS145: Experiments in Computer Organization

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
2.0	2.4	8 weeks	lecture: 120

The main purpose of this course is to deepen students' understanding of concepts and principles in the computer system structure. The course teaches students the hardware

description languages Verilog, and guides them step by step to build an ALU, a memory unit and finally a CPU. The CPU should support basic MIPS instruction set and run in a pipelined manner. Students are also encouraged to devise schemes (e.g. prefetching and predictions) to prevent stalls of the pipeline. The design will be simulated in a developing environment and eventually downloaded into FPGAs for verification.

CS339: Computer Network

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

Computer Networks is the core course for undergraduate students in department of computer science. This introductory course focuses on explaining how the network system works, ranging from physical layer to application-level protocols. It also explains the principles of how to design networks and network protocols. Students gain experience by reading recent research papers to understand what a modern network system should do. The course grounds many of the concepts in current practice and recent developments, such as software defined networks, data centers, and Internet of Things.

The course introduces concepts and developments in today's computer networks. In each layer of network, a variety of practical examples are presented. Students will comprehend the first principles in widely-deployed networks such as Ethernet, WiFi, SONET, and protocols such as HTTP and SMTP.

Textbook: *Computer Networks*, by Andrew S. Tanenbaum.

CS307: Operating Systems

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

An Operating System is a complex software package that manages the resources of a computer system, and provides the base upon which applications can be written. In this course, we will study the basic components of an operating system, their functions, mechanisms, policies and techniques used in their implementation and several examples from popular operating systems. Practical projects using experimental operating systems will support concepts discussed during the lecture. On completion of this course, students will be able to understand the components and working of an operating system and to design and implement various operating system functions.

Contents include system calls, process and threads, scheduling algorithm, memory management, virtual memory and page fault handling, file system.

Textbook: *Operating System Concepts*, by A. Silberschatz, P. Galvin, G. Gagne.

CS356: Project Workshop of Operating Systems

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
2.0	2.4	16 weeks	lecture: 90

Operating system projects are closely connected with its main course, CS307, and combines theory with practice. With open source code analysis,and kernel and application

module design, the course enables the students to further understand the design principles of operating system through practice, and to master the key implementation technology.

EE204: Embedded System and Microcomputer Principles

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
4.0	4.8	16 weeks	lecture and lab: 180

Through classroom teaching and hands-on experiments, the course enables students to master the principles of computer memory, CPU, internal structure, the general purpose I/O interface, instruction system and assembly language. It also introduces several peripheral chips (e.g. timer, counter, 7-segment digit driver) and communication protocols (e.g. I^2C, UART) that are broadly used in embedded systems.

The course is divided into two parts. The first part teaches basic concepts using the classical 80x06 architecture as a running example. Students are required to code in assembly language and simulate their codes in a virtual environment. The second part is experiment-based, teaching students to implement more advanced tasks (e.g. communicating with peripherals) on a real Cortex-M3/M4 board. Students will be asked to read datasheets themselves and figure out a solution. Students are graded by completion of their assignments, experiments, the scores of a theory exam and an experiment exam.

Textbook: *Computer Organization and Architecture*, by William Stalling.

IS312: Information Theory and Coding Technology

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
2.0	2.4	16 weeks	lecture: 90

The aim of the course is to teach the basic knowledge of Shannon information theory and communication coding technology for inter-disciplinary students in the research fields of computer science, communication, information security and applied mathematics. The course consists of two parts. The first part is of information theory, which includes the measurement of information (entropy, relative entropy, mutual information); weakly typical sequence (for data compression) and Shannon Source Theorem. The second part is of coding technology, which includes source coding (Huffman Code, arithmetic code, Lempel-Ziv Algorithm, Run Length Code, JPEG), and also channel coding (linear codes, cyclic codes, Hamming codes, RS codes, LDPC code, decoding principles etc). Furthermore, some basic knowledge about finite field and probability theory will be reviewed.

Textbook: *Information Theory and Network Coding*, by Raymond W. Yeung.

IS307: Introduction to Cryptography and Information Security

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

This course is for 3rd year undergraduates in Computer science. It introduces the basic concepts in cryptography and information security, including classical and modern ciphers, hash functions, and security protocols such as Kerberos, SSL/TLS, IPSEC, PGP, and computer and network related security issues such as authentication, web security, malware, attacks and preventions.

Textbook: *Cryptography and Network Security: Principles and Practice*, by William Stalling.

CS383: Programming Languages

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

Only a small number of concepts underlie the hundreds of programming languages that have been designed and implemented. This course introduces the concepts that serve as a basis for programming languages. It aims to provide the students with a solid understanding and appreciation of the various essential programming language constructs, programming paradigms, evaluation criteria and language implementation issues. The course covers concepts from functional, imperative, object-oriented, logic programming and optionally scripting languages. These concepts are demonstrated by incrementally adding more features to a simple functional language, the lambda calculus, to eventually arrive at a full-fledged, user-friendly general purpose programming language. At the end of the course, a number of modern programming languages from various paradigms are presented as examples.

Example languages used in this course: C/C++ for imperative programming; Java and SmallTalk for object-oriented programming; ML, Haskell and Lisp for functional programming; Prolog for logic programming.

Types of assignments: (1) Programming assignments in all of the four paradigms. (2) Theoretical proofs of lanaguage properties and safety. (3) Implementation of an interpreter.

Textbooks:

- *Types and Programming Languages*, by Benjamin C. Pierce.
- *Practical Foundations for Programming Languages*, by Robert Harper.

SE353: Program Analysis and Verification

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

This course teaches students program analysis and program verification techniques, enabling them to understand whether a program has correctly achieved its intended goals and to use mathematics to rigorously prove that a program meets its intended goals.

Main teaching contents: program analysis introduction, data flow analysis technology, data flow analysis framework, program analysis tool, program debugging, program verification introduction, TLA$^+$ specification and verification tool, model checking, Hoare logic and automated generation of loop invariants.

Textbooks:

- *Compilers – Principles, Techniques, and Tools*, by Alfred V. Aho, Monica S. Lam, Ravi Sethi and Jeffery D. Ullman.
- *The Formal Semantics of Programming Languages*, by Glynn Winskel.

CS445: Combinatorics

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
3.0	3.6	16 weeks	lecture: 135

Combinatorics is a vast field of study, and this course focuses only on graph theory. We will first recapitulate relevant knowledge in discrete mathematics courses, and then proceed to deeper results in graph theory. The course intends to convey some important classical results in graph theory, and to cultivate the mode of thinking in graph theory. It also stresses applications in computer science.

Contents include: matching in bipartite and general graphs, weighted matching and weighted vertex cover, chromatic number and chromatic polynomial, Brook's Theorem and Five Color Theorem, planar graphs, flows, Ramsey theory.

Textbook: *Graph Theory*, by Reinhard Diestel.

CS431: Professional Practice

Credits	*ECTS Equivalent*	*Duration*	*Length (minutes per week)*
2.0	2.4	4 weeks	full-time

This course requires students to practice in a lab or an enterprise, where students can make use of the knowledge learned before in the research/production process, and get more emotional contact to the actual working environment. This course is an indispensable and important part of the education and teaching system. It is not only a continuation of the school teaching, but also a summary of the school teaching.

The department offers a variety of topics/projects for students who are interested in doing research in academia. Students can report their interests to the department for selection. Each topic/project is led by a supervisor expertised in the area. During the whole month, students are expected to participate full-time. At the end of the process, students will submit a report summarizing their work.